

Mobile Radiation

I am a regular reader of EFY and eagerly wait for the magazine every month. I love the articles published in EFY. The "How To Avoid Mobile Radiation" article published in Dec. 2011 issue was an eye-opener for mobile users. The facts and figures helped readers to digest the concept.

I suggest you to publish some home-made projects that can be easily assembled even by novices.

D. Santhosh Kumar
K.L. University, Vijayawada,
Andhra Pradesh

Robotics

I am very happy to see the articles 'Robot That Breathes Life' and 'Kits That Maketh Robots' published in March issue. The articles contained a lot of useful information. Thank you EFY!

Ashwin Kavalayil
Through e-mail

EFY Circuits

Thank you for the courtesy extended during EFY Exhibition! EFY has improved a lot since the time I personally assembled a circuit some 25 years ago. I suggest following modifications in the circuits published in Dec. 2011 issue:

1. **Mains Monitor.** In this circuit, I would like to add a feature that predicts the mains load-level condition. With this feature, the user can avoid loading during load on/off-peak time, saving energy in the process.

2. **Single-Control Switch for Fan and AC.** Here, I prefer to have the AC and fan operated sequentially so that when the AC is 'on' the fan is 'off', and vice versa—based on a time interval fixed by a potmeter/switch, on repetitive logic.

N. Dhakshina Moorthi
Sr Manager-Development,
Sree Vishnu Magnetics,

Kancheepuram Distt, Tamil Nadu
EFY: Thanks for the feedback! We have forwarded your suggestions to the authors of these circuits. We will forward their comments to you on receipt.

Robotic Arm

For the 'Robotic Arm' construction project published in Nov. 2011 issue, I need information on the kind of controller used, maximum speed of the arm, maximum weight it can lift and standard robotic arm material.

Ajeet Samani

Kolhapur, Maharashtra

The author Venkat Raju replies: PIC18F4431 microcontroller is used as mentioned in the text. The maximum speed depends on the type of servo motor used and the input voltage. The maximum weight depends on the mechanical structure and torque distribution of the motors. For more information, please refer to the formula given in the text. The robotic arm material requirement depends on the load and the application. For low-end applications, aluminium will suffice.

Holi e-Card

I want to compile the source code of 'Holi e-Card in C' program, published in EFY March, in Linux operating system. How can I execute that file?

Sekhar
Hyderabad

The author Nishant Mittal replies:

In the case of Linux, the .bgi path for the graphics is different. The user has to assign the graphics path appropriately in order to execute the graphics file.

First, run 'sudo apt-get build-essential' to install necessary compiler tools. Then, install the following packages:

```
libSDL-image1.2
libSDL-image1.2-dev
guile-1.8
guile-1.8-dev
libSDL1.2debian-arts
libarts0-dev
libaudiofile-dev
libesd0-dev
libdirectfb-dev
libdirectfb-extra
libfreetype6-dev
libxext-dev
x11proto-xext-dev
libfreetype6 (upgrade)
libaa1
```

```
libaa1-dev
libslang2-dev
libasound2
libasound-dev
```

You can install these via Synaptic Package Manager one by one. (To open Synaptic Package Manager, go to System→Administration→Synaptic Package Manager.) Or, type the following command in the terminal window:

```
sudo apt-get install libSDL-image1.2,
libSDL-image1.2-dev, guile-1.8,
guile-1.8-dev, libSDL1.2debian-arts,
libarts0-dev, libaudiofile-dev,
libesd0-dev, libdirectfb-dev,
libdirectfb-extra, libfreetype6-dev,
libxext-dev, x11proto-xext-dev,
libfreetype6 (upgrade), libaa1, li-
baal-dev, libslang2-dev, libasound2,
libasound-dev
```

Thereafter, download 'libgraph-1.0.1' from 'http://download.savannah.gnu.org/releases/libgraph/' to your home folder. Right-click this file and press 'Extract here' option.

Now type following commands in the terminal window:

```
./configure
sudo make
sudo make install
```

Add the following lines in your source program:

```
int gd, gm=VGA_MAX; gd=DETECT;
initgraph(&gd, &gm, &gm, &gm, &gm, &gm);
Then compile using g++ as follows:
g++ test.cpp -o test.o -lgraph
```

Execute the program using the command:

```
./test.o
```

If compilation error says "could not load shared libraries" or something like that, just run the following command to fix it:

```
sudo cp /usr/local/lib/libgraph.*
/usr/lib
```

Laptop Audio Amplifier

In 'Laptop Audio Amplifier' circuit published in February issue, the correct value for C8 should be 100 μ F (not 330 μ F as shown in the published circuit).

S. Aravind
The author